

GPIE — Green Policy Intelligence Engine

Independently Verifying Environmental Policy Claims Using Satellite Data

Executive Summary · DOI: 10.5281/zenodo.21756661 · Sakshi D. Maske

The Question

Did the European Union's Climate Law (effective 30 June 2021) produce a measurable, statistically distinguishable reduction in NO₂ air pollution across the EU-27 — beyond what would have happened anyway? Governments report their own climate progress; that self-reported progress is rarely audited against independent, physical evidence. GPIE builds that independent audit layer using satellite data rather than assuming the policy worked.

The Method

A two-group Difference-in-Differences (DiD) design compares EU-27 countries against a genuine non-EU control group — the UK, Norway, and Switzerland — across 8 independently-sourced datasets (NO₂ via Sentinel-5P TROPOMI, NDVI via CGLS, climate via ERA5, GDP via Eurostat/World Bank, land cover, elevation, administrative boundaries, and EU policy records), spanning 30 countries and 2019–2024. Standard errors are clustered by country throughout.

The Finding

No statistically distinguishable EU-specific reduction in NO₂ was detected once genuinely compared against a non-EU control group — independently confirmed via a 23-quarter event-study and five additional robustness checks. Reporting this honest null result, rather than the flawed 'significant' result an earlier single-cohort design produced, is itself the project's core finding.

Metric	NO2 (Primary)	NDVI (Secondary)
DiD Coefficient	-1.40e-6	-0.0210
P-value (cluster-robust)	0.663 — not significant	0.012 — significant
95% Confidence Interval	[-7.68e-6, +4.88e-6]	[-0.0372, -0.0047]

The NDVI result is reported as an honest, exploratory secondary finding meriting further investigation — not as evidence the Climate Law itself affected vegetation health, since this analysis does not control for land-use change or drought-driven vegetation stress.

Validation & Robustness Checklist

- ✓ Cluster-robust standard errors, clustered by country (Bertrand, Duflo & Mullainathan, 2004)
- ✓ Genuine external control group — 3 non-EU countries (UK, Norway, Switzerland)
- ✓ Placebo test — caught and fixed a flawed initial single-cohort design (which had wrongly shown significance)
- ✓ 23-quarter event-study validation — confirms parallel pre-trends and no hidden delayed effect
- ✓ 5 additional robustness checks — GDP removed, log-transformed outcome, treatment-date shifted $\pm 6 \pm 12$ months, pollution-level subgroup split, and a formal minimum-detectable-effect calculation
- ✓ Honest null result reported — not forced or reframed as significant

Honest Limitation

The control group consists of only 3 countries. At 80% statistical power, this design can reliably detect an effect of roughly 28% of baseline EU NO₂ or larger — this rules out an effect of that size, but not a smaller true effect. The honest conclusion is not “the policy had no effect,” but that no effect of at least ~28% could be detected with this sample and design.

Global Transferability

The NO₂ acquisition pipeline was separately tested, with zero modification to the core code, on India (2019–2024) — returning physically realistic values consistent with the EU-27 dataset's observed range. This confirms the framework's architecture is portable beyond a single region, not a one-country tool.

GitHub: github.com/sakshimaske303-commits/GPIE | **Live Dashboard:** f5cf6fijj9gm564r6aapt6.streamlit.app | **Zenodo DOI:** 10.5281/zenodo.21756661

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